

National University of Computer and Emerging Sciences, Lahore Campus



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Type:	Announced	Page(s):	2
Exam:	Quiz 4A Solution	Section:	

Name & Roll No:

Question # 1:

[10 marks, CLO # 2]

Write the output for the following piece of code assuming that there are 4 MPI processes.
Assume there is no syntax error.

```
int main(int argc, char** argv) {
    MPI_Init(&argc, &argv);
    MPI_Status status;
    int p, i, b;
    MPI_Comm_size(MPI_COMM_WORLD, &p);
    int my_rank;
    MPI_Comm_rank(MPI_COMM_WORLD, &my_rank);
    int a=my_rank;
    int sendTag=a;
    int recvTag=1;
    int next=(my_rank+1)%p; //determine my right node
    int previous=((my_rank-1+p)%p); //determine my left node
    MPI_Sendrecv(&a, 1, MPI_INT, next, sendTag, &b, 1, MPI_INT, previous, recvTag,
MPI_COMM_WORLD, &status);
    printf("I\'m %d: Received:%d from %d and Sent:%d to %d\n ",
my_rank , b, previous, a, next);
    MPI_Finalize();
}
```

OUTPUT:

Output (Solution):

I\'m 0: Received:3 from 3 and Sent:0 to 1

I\'m 1: Received:0 from 0 and Sent:1 to 2

I\'m 2: Received:1 from 1 and Sent:2 to 3

I\'m 3: Received:2 from 2 and Sent:3 to 0

Question # 2:**[6 marks, CLO # 3]**

(i) Assume these are the contents of sbuf in the processes P0, P1 and P2:

P0	2	3	1	7	11	0
P1	6	3	6	2	8	12
P2	2	4	4	10	4	5

Show the contents of rbuf in the same processes after this MPI command:

`MPI_Scan(sbuf, rbuf, 6, MPI_INT, MPI_SUM, MPI_COMM_WORLD);`

Hint: (MPI_Scan is used for prefix-sum).

Contents of rbuf:

<i>P0</i>	<i>2</i>	<i>3</i>	<i>1</i>	<i>7</i>	<i>11</i>	<i>0</i>
<i>P1</i>	<i>8</i>	<i>6</i>	<i>7</i>	<i>9</i>	<i>19</i>	<i>12</i>
<i>P2</i>	<i>10</i>	<i>10</i>	<i>11</i>	<i>19</i>	<i>23</i>	<i>17</i>